Introduction to Software Development

Spring, 2022

Homework two

Instructor Wei Fang

READ INSTRUCTIONS AND FOLLOW CAREFULLY!

Assignment two of three (5% towards your final course grade)

Instructions

This assignment is due promptly by 11:59:59 PM on Wednesday March 23. We will go over answers during the lecture on March 24.

Please type your answers clearly and carefully.

You need to submit this file to the course's page on Canvas. No email or print submission will be accepted.

For question 1, insert snapshots of results of your working programs.

For question 2 and 3, make sure you insert your source codes and snapshots of results.

For question4, write one single program. Insert your source codes and snapshot of result.

This is a Word document. Insert your answers right below the questions. Submission in other formats will result deduction of one point.

Please make sure that you submit correct version. If you submitted wrong version, only the version closes to the deadline will be graded and counted.

Question 1: Copy and paste screen shots of your program and result screen shots. What would the following program print out? Explain briefly in English about each and every output (1 point)

```
void f(const int a = 5)
{
    std::cout << a*2 << "\n";
}

int a = 123;
int main()
{
    f(1);
    f(a);
    int b = 3;
    f(b);
    int a = 4;
    f(a);
    f();
}
```

//Answer here:

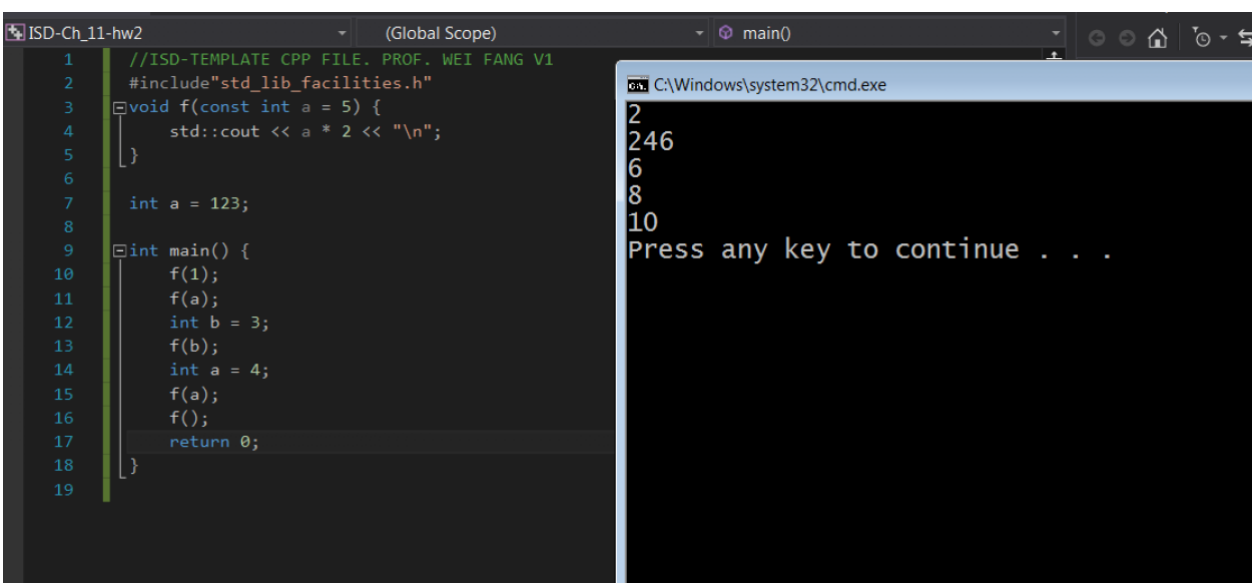
f(1): we are giving 1 to the function so the output is 2.

f(a): we have set up a global variable a = 123 so the output is 246;

f(b): we have declare a new variable and use it as input of function so the output is 6

f(a): since we have been declare a new local variable in the scope and give the value of 4 so the output is 8

f(): we define a int constant of 5 when writing the function so the output is 10 when we not identify an input.

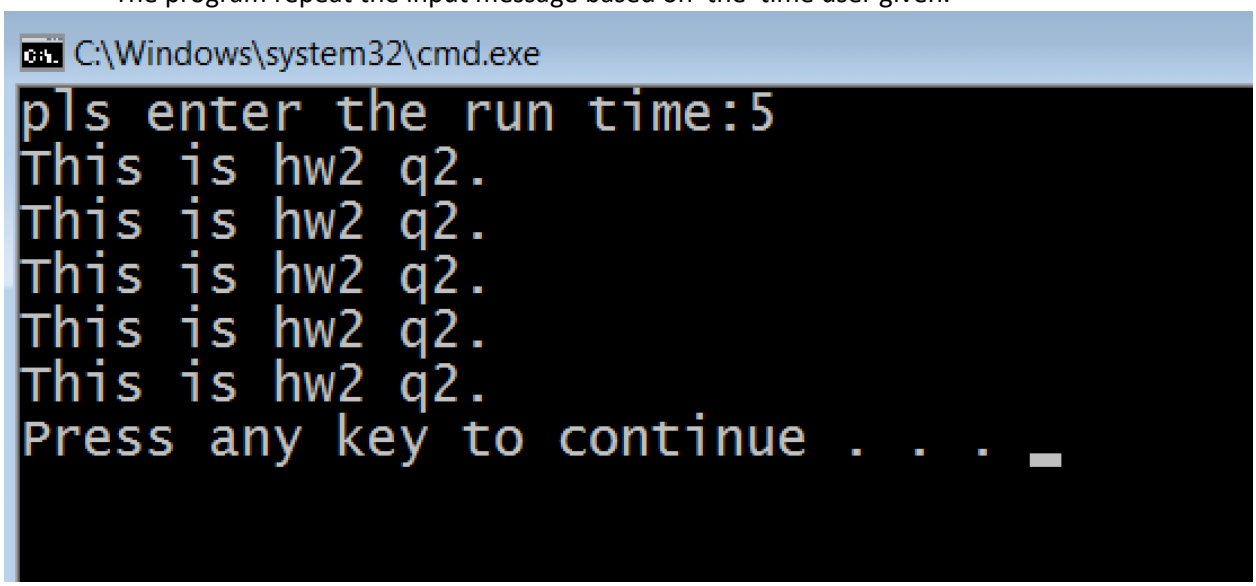


Question 2: c/p screen shots of your entire program and results. What does the function below do? **Explain** briefly in English and **finish** the whole program by allowing the user to input in the run time. Draw a flowchart (1 points)

```
void test (char *test = "\n", int n = 1 ) {
        for ( int i = 0; i < n; ++i ) {
                cout <<  test;
        }
}
```

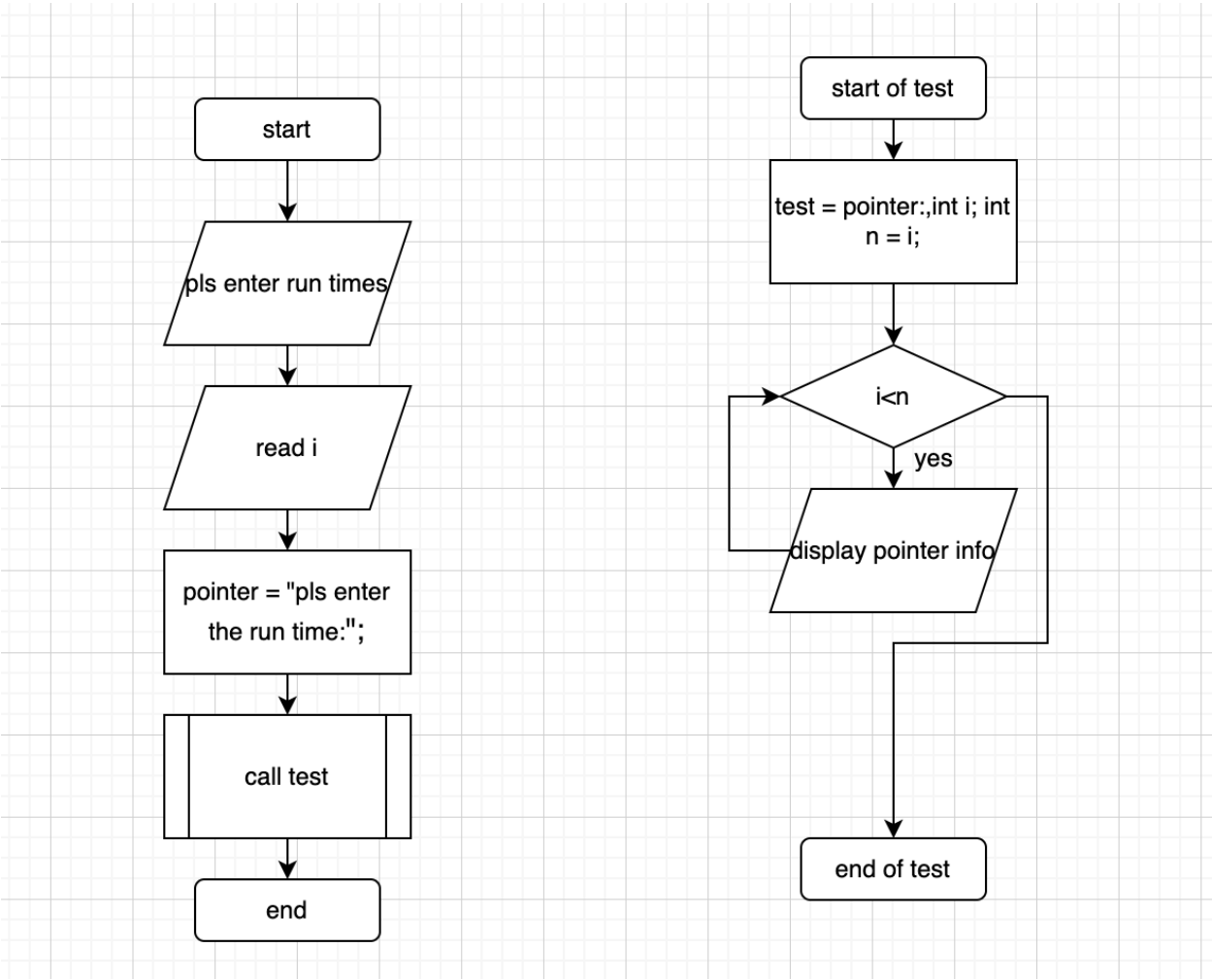
//your program and working snapshot here:

The program repeat the input message based on the time user given.




```
#include "std_lib_facilities.h"
void test(char *test ="\n", int n = 1) {

        for (int i = 0; i < n; ++i) {

                cout << test;

        }

}

int main() {
        int i;
        cout << "pls enter the run time:";
        cin >> i;
```

```
        char *pointer = "This is hw2 q2.\n";
        test(pointer, i);
}
```



Question 3: copy and paste screen shots of your entire program and results. Consider the following function, finish the function by allowing it to identify if x is a negative number or a nonnegative number. Write a **full program using this function** to output the result to the user. (1 points)

```
void setTest (char **setPointer) {
        int x;
        cin >> x;
        // your function code here
        cout<<"\n";
        cout<<"Homework 2 q3.";
}
```

//your main program and working snapshot here

```
//ISD-TEMPLATE CPP FILE. PROF. WEI FANG V1
#include"std_lib_facilities.h"

void setTest(char **setPointer) {

        int x;
        cout << "input number";
        cin >> x;
        // your function code here
        if (x > 0) {
                cout << "positive\n";
        }
        else {
                cout << "negative\n";
        }
        cout << "\n";
        cout << "Homework 2 q3.";
}


int main() {
        char*test;
        setTest(&test);
        return 0;
}
```

C:\Windows\system32\cmd.exe

```
input number5
positive

Homework 2 q3.
Press any key to continue . . .
```

C:\Windows\system32\cmd.exe

```
input number-5
negative

Homework 2 q3.
Press any key to continue . . .
```

Question 4: copy and paste screen shots of your entire program and results. Write a program that takes nine integer as its arguments from command line. The first eight arguments are going to be sorted using the bubble sort. The ninth numeric argument is going to be used as the keyword that will be searched for by your program using binary search. Your program should display the original unsorted inputs (note: without the 9th argument, which is the searching keyword), then displays the sorted inputs, and then displays the position of the keyword in the sorted array. Your program should be able to handle basic exceptions as demoed below (2 points):

```
C:\ >ISD-Homework2-Q4.exe
Please type in at least one argument.
C:\ >ISD-Homework2-Q4.exe 23 4 3 5 6
Pleas type in exact 9 arguments.
C:\ >ISD-Homework2-Q4.exe 5 6 9 10 94 2 5 3 10
Your original inputs are:
5
6
9
10
94
2
5
3
Sorted inputs are:
2 in position: 0
3 in position: 1
5 in position: 2
5 in position: 3
6 in position: 4
9 in position: 5
10 in position: 6
94 in position: 7

That number 10 is found at element 6 in the array.
```

//Your codes and working snapshot below

```
C:\Users\c\Documents\Visual Studio 2015\Projects\ISD-Ch_11-hw2\Debug>ISD-Ch_11-h
w2.exe
pls enter 9 number
C:\Users\c\Documents\Visual Studio 2015\Projects\ISD-Ch_11-hw2\Debug>ISD-Ch_11-h
w2.exe 2 3
pls enter exactly 9 numbers
C:\Users\c\Documents\Visual Studio 2015\Projects\ISD-Ch_11-hw2\Debug>ISD-Ch_11-h
w2.exe 5 6 9 10 94 2 5 3 10
unsorted:  5  6  9  10  94  2  5  3
sorted:  2  3  5  5  6  9  10  94
The number of 10is found at position 6

C:\Users\c\Documents\Visual Studio 2015\Projects\ISD-Ch_11-hw2\Debug>_
```

```
//ISD-TEMPLATE CPP FILE. PROF. WEI FANG V1
#include"std_lib_facilities.h"

void bubbleSort(int array[], int size) {
	for (int step = 0; step < (size - 1); ++step) {

		int swapped = 0;
		for (int i = 0; i < (size - step - 1); ++i) {
			if (array[i] > array[i + 1]) {
				int temp = array[i];
				array[i] = array[i + 1];
				array[i + 1] = temp;

				swapped = 1;
			}
		}
		if (swapped == 0)
			break;
	}
}

int binarySearch(int array[], int x, int low, int high) {
	while (low <= high) {
		int mid = low + (high - low) / 2;
		if (array[mid] == x)
			return mid;
		if (array[mid] < x)
			low = mid + 1;
		else
			high = mid - 1;
	}
	return -1;
}
```

```
void printArray(int array[], int size) {
	for (int i = 0; i < size; ++i) {
		cout << "  " << array[i];
	}
	cout << "\n";
}


int main(int argc,char*argv[]) {
	int numbers[8];
	int data[] = { -2, 45, 0, 11, -9 };
	int size = sizeof(data) / sizeof(data[0]);

	if (argc == 1) { cout << "pls enter 9 number"; }
	else if (argc != 10) { cout << "pls enter exactly 9 numbers"; }
	else {
		for (int i = 0; i < 8; i++) {
			numbers[i] = atof(argv[i+1]);
		}
		cout << "unsorted:";
		printArray(numbers, 8);
		bubbleSort(numbers, 8);
		cout << "sorted:";
		printArray(numbers, 8);
		cout << "The number of " << atof(argv[9]) << "is found at position "
			<< binarySearch(numbers, atof(argv[9]), 0, 8) << "\n";

	}
	return 0;
}
```